

Ordinary Differential Equations



Prof. Alex Chen

Department of Physics

Washington University in St. Louis

Today



- **The problem.** ODEs and initial value problems.
- **Euler's method.** The simplest solver and its error.
- **Runge-Kutta methods.** The midpoint method and RK4.
- **Convergence.** Checking that a numerical solution can be trusted.

Part I · The problem



ODEs · Initial value problems

Ordinary differential equations



An ODE relates an unknown function $y(t)$ to its derivative:

$$\frac{dy}{dt} = f(t, y)$$

- Example: $dy/dt = -y$ has the solutions $y(t) = Ce^{-t}$. An ODE alone never fixes the constant; some extra condition has to.
- y can be a vector $\mathbf{y}$: several unknown functions, coupled through one vector function $\vec{f}$.

Newton's equations fit this form



Physics mostly hands us *second* derivatives. A pendulum:

$$\ddot{\theta} = -\frac{g}{L}\sin\theta$$

Define $\omega = \dot{\theta}$ and it becomes a first-order system:

$$\mathbf{y} = \begin{pmatrix} \theta \\ \omega \end{pmatrix}, \quad \frac{d\mathbf{y}}{dt} = \vec{f}(\mathbf{y}) = \begin{pmatrix} \omega \\ -\frac{g}{L}\sin\theta \end{pmatrix}$$

Any n -th order ODE becomes a system of n first-order ODEs this way. Solvers only ever need to handle $d\mathbf{y}/dt = \vec{f}(t, \mathbf{y})$.

Initial value problems



An *initial value problem* (IVP) supplies the missing condition at one point:

$$\frac{dy}{dt} = f(t, y), \quad y(t_0) = y_0$$

- This is time evolution: the state of the system now, predict it later. Most of mechanics, circuits, orbits, and population models take this form.
- The other common type is the *boundary value problem*: conditions at both ends of an interval, such as a field with fixed values on two plates. IVPs now; BVPs later in the course.

Part II · Euler's method

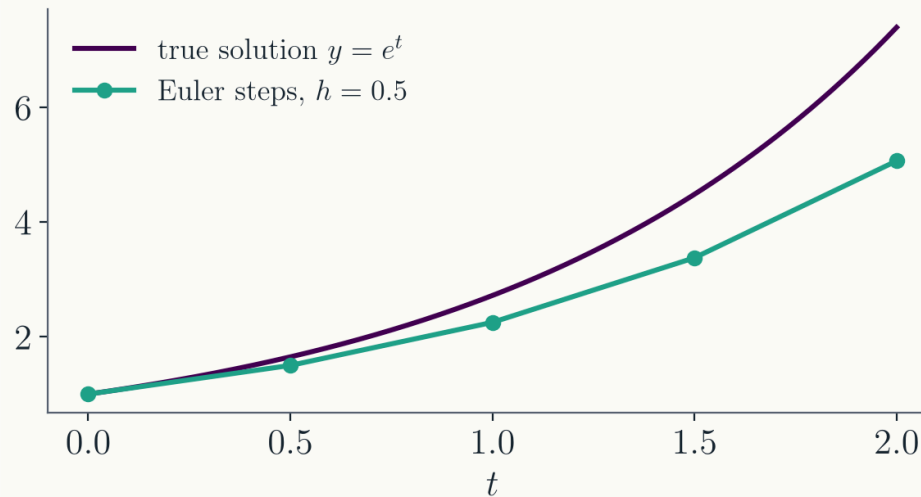


Follow the tangent

Euler's method

The ODE gives the slope at every point. Step h along the current slope and repeat:

$$y_{n+1} = y_n + h f(t_n, y_n)$$



Each step is exact for a straight line, so curvature is what it misses; here every step lands below the curving-up exponential.

Euler's method: the error

- One step is a Taylor series cut after the first derivative:

$$y(t + h) = y(t) + h y'(t) + O(h^2)$$

so the error of a single step is $O(h^2)$.

- Reaching a fixed time T takes $N = T/h$ steps. As with the integration rules: per-step error, times N , loses one power of h . The accumulated error is $O(h)$.
- First order: halve h , halve the error, and double the work.

Drawbacks of Euler's method



- First order is expensive: every digit of accuracy costs a factor of 10 in steps.
- The error compounds: each step starts from the slightly wrong point the previous step produced.
- It can be unstable: for some equations the numerical solution blows up even though the true solution decays. Defined and analyzed next lecture.

Almost nobody should use Euler as-is.

Part III · Runge-Kutta methods



Better slopes, same idea

The midpoint method



Use the Euler step only to *estimate* the slope at the middle of the interval, then take the real step with that slope:

$$k_1 = f(t_n, y_n)$$

$$k_2 = f\left(t_n + \frac{h}{2}, y_n + \frac{h}{2}k_1\right)$$

$$y_{n+1} = y_n + h k_2$$

Two evaluations of f per step instead of one. What does it buy?

Error of the midpoint method



Expand $y(t + h)$ and $y(t)$ around the midpoint, writing $y_m = y(t + h/2)$:

$$\begin{aligned}y(t + h) &= y_m + \frac{h}{2} y'_m + \frac{h^2}{8} y''_m + O(h^3) \\y(t) &= y_m - \frac{h}{2} y'_m + \frac{h^2}{8} y''_m + O(h^3)\end{aligned}$$

Subtract: the y'' terms cancel by symmetry,

$$y(t + h) = y(t) + h y' \left(t + \frac{h}{2} \right) + O(h^3)$$

What is the cumulative error?

Stepping with the midpoint slope is exact through h^2 . Same symmetry cancellation as the trapezoid error in Lecture 5.

Fourth-order Runge-Kutta



The same idea pushed further: sample four slopes per step, average them with fixed weights. The midpoint method is called RK2; this is *RK4*:

$$k_1 = f(t_n, y_n)$$

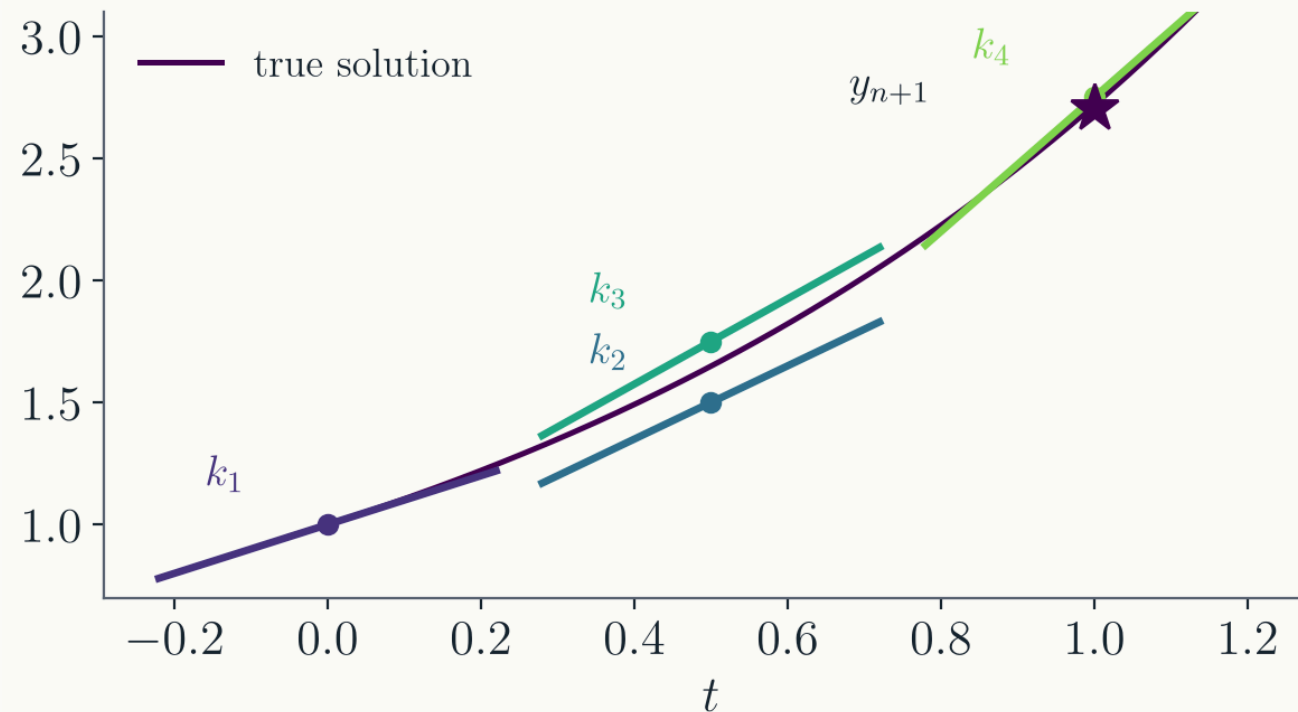
$$k_2 = f\left(t_n + \frac{h}{2}, y_n + \frac{h}{2}k_1\right)$$

$$k_3 = f\left(t_n + \frac{h}{2}, y_n + \frac{h}{2}k_2\right)$$

$$k_4 = f(t_n + h, y_n + h k_3)$$

$$y_{n+1} = y_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

One RK4 step



- k_1 : slope at the start. k_2, k_3 : two successively better midpoint slopes. k_4 : slope at the far end.
- On $y' = y$ with $h = 1$, one step gives $y_1 = 65/24 = 2.7083$, against $e = 2.7183$: the step reproduces the Taylor series of the solution through h^4 .

RK4 facts



- Fourth order: local error $O(h^5)$, accumulated error $O(h^4)$. Halve h , gain a factor of 16.
- If f depends only on t , the ODE is an integral, and RK4 reduces exactly to Simpson's rule: the weights $\frac{1}{6}, \frac{1}{3}, \frac{1}{3}, \frac{1}{6}$ are Simpson's 1, 4, 1 over a double interval, since both k_2 and k_3 are evaluated at the midpoint.

For much of physics, RK4 is the default solver. In many papers "Runge-Kutta" means RK4.

Drawbacks of RK4



- The step size is fixed: h has to be chosen for the most difficult part of the solution, and everywhere else the work is wasted.
- It is not built for *stiff* equations, where stability rather than accuracy limits the step (next lecture).
- Sometimes fourth order is not enough; long integrations of orbits want higher order or structure preservation.

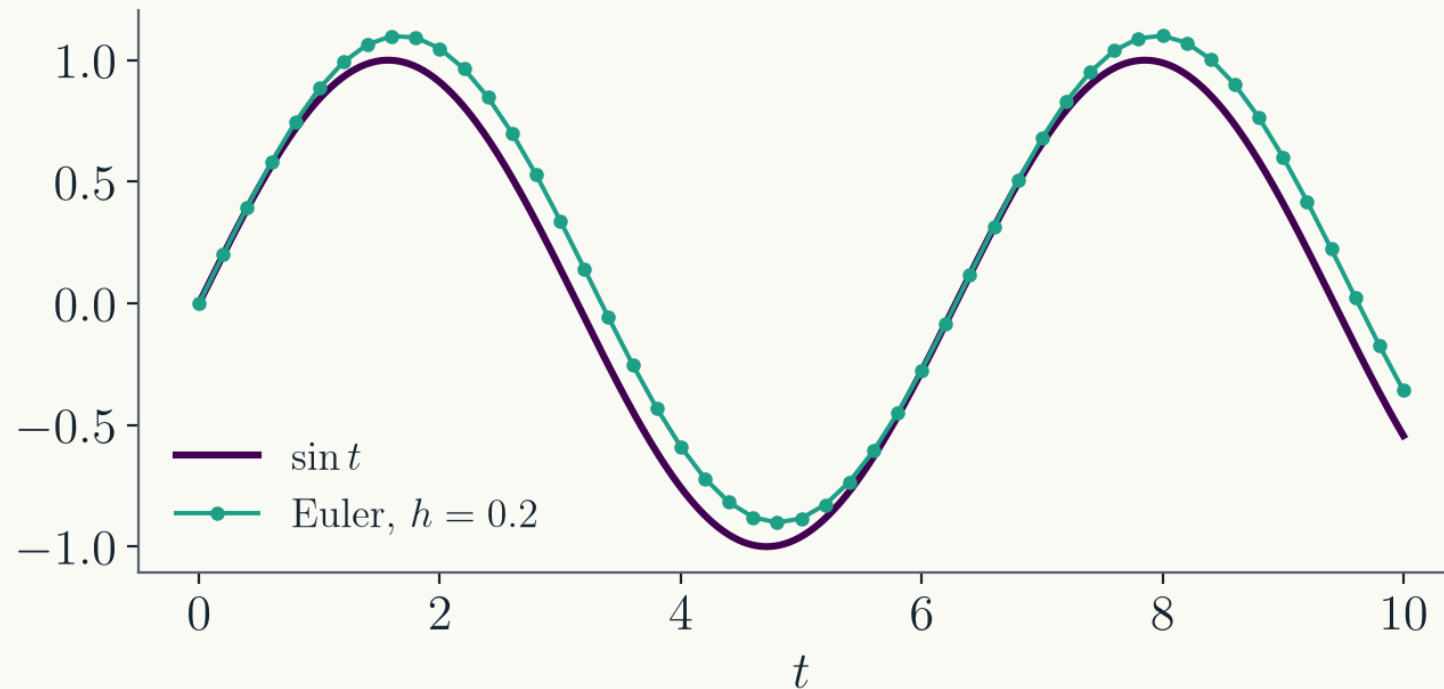
Part IV · Does it converge?



Estimate, then measure

A test problem

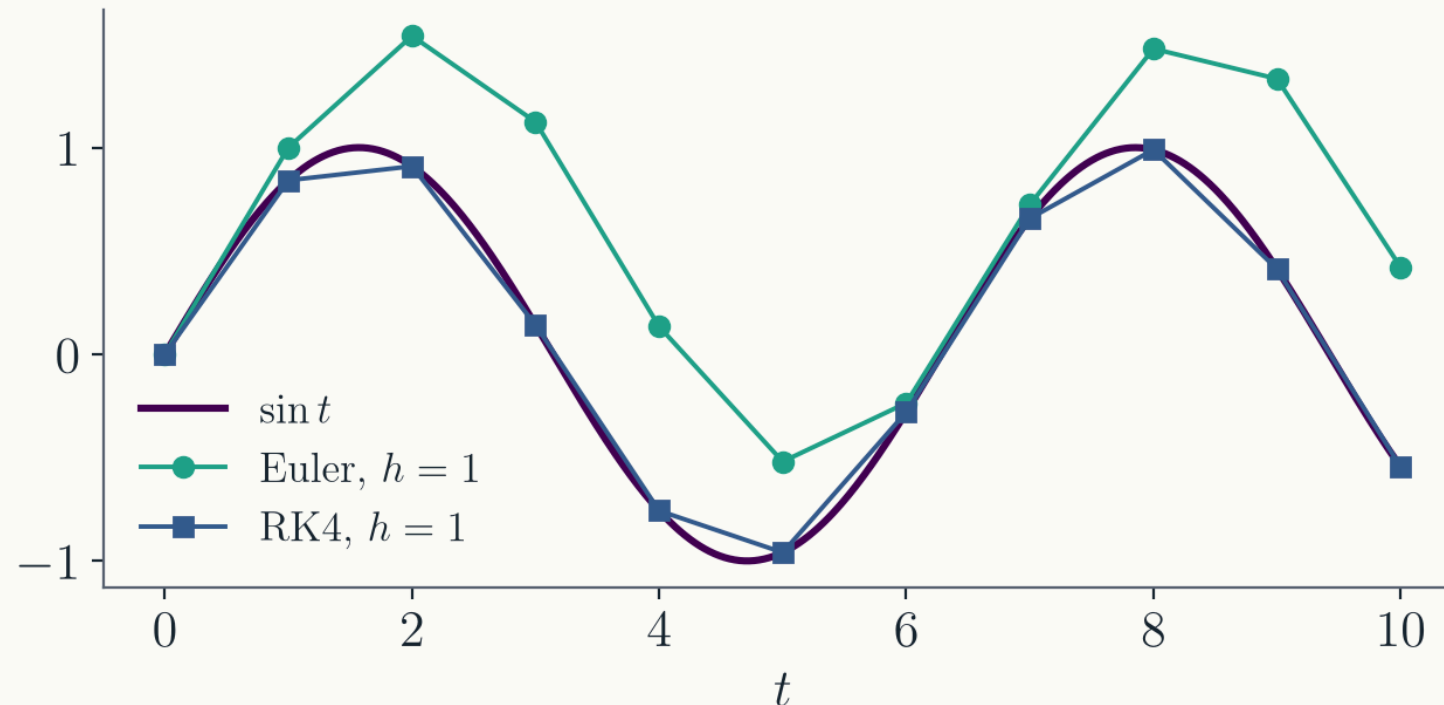
$dy/dt = \cos t$ with $y(0) = 0$, whose exact solution is $y = \sin t$. Euler with $h = 0.2$:



The drift reaches 0.2 by $t = 10$: first order means the error is about h , not about machine precision.

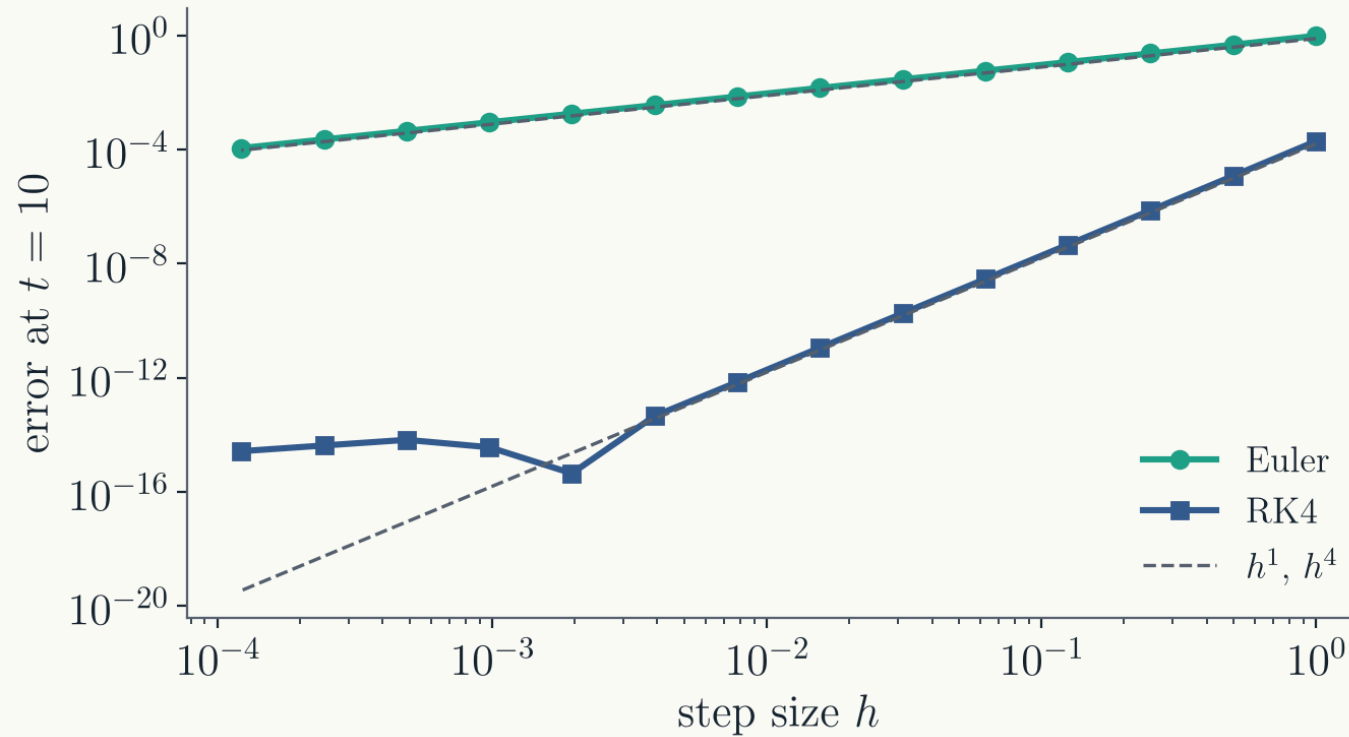
RK4 comparison

Both methods with the very coarse step $h = 1$:



- Euler is off by order 1; RK4 is off by 3.5×10^{-4} , with only ten steps and forty evaluations of f .

Measured convergence



- Measured slopes: 1.006 and 4.005.
- Below $h \approx 10^{-3}$, RK4 leaves its line and flattens near 10^{-15} : the rounding floor. Smaller steps buy nothing more.

Without an exact solution

- The test problem had $\sin t$ to compare against. In a real problem we do not know the solution in advance.
- The practical check: solve at h and at $h/2$, and compare. If the two agree to your tolerance, the answer can be trusted at that level; if not, refine.
- This is the same move as the doubling trapezoid in Lecture 5, and it can be automated into a solver that picks its own step size. That is where the next lecture ends up.

Where this goes next



- **Next lecture:** where the RK coefficients come from, stiff equations and stability, and implicit methods.
- The ODE machinery culminates in Project 1: tracing the image of a black hole.
- References:
 - Numerical Recipes, 3rd ed., Chapter 17 (ODEs).
 - Burden & Faires, *Numerical Analysis*, Chapter 5.